EXPERIMENT NO 19

Determination and plotting of Antenna Efficiency.

Aim:

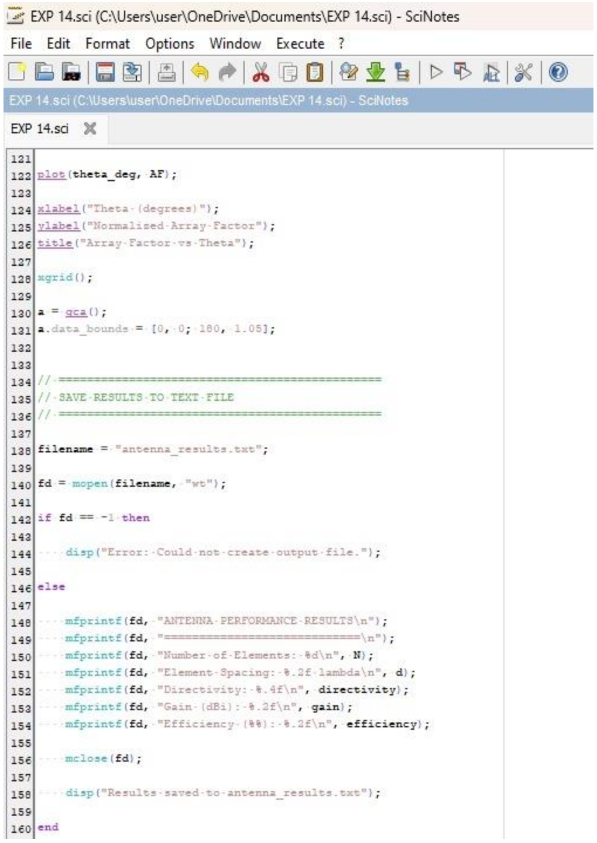
To determine and plot Antenna Efficiency and to Estimate the efficiency based on radiated and input power.

Software required:

- SCILAB version 6.0.1

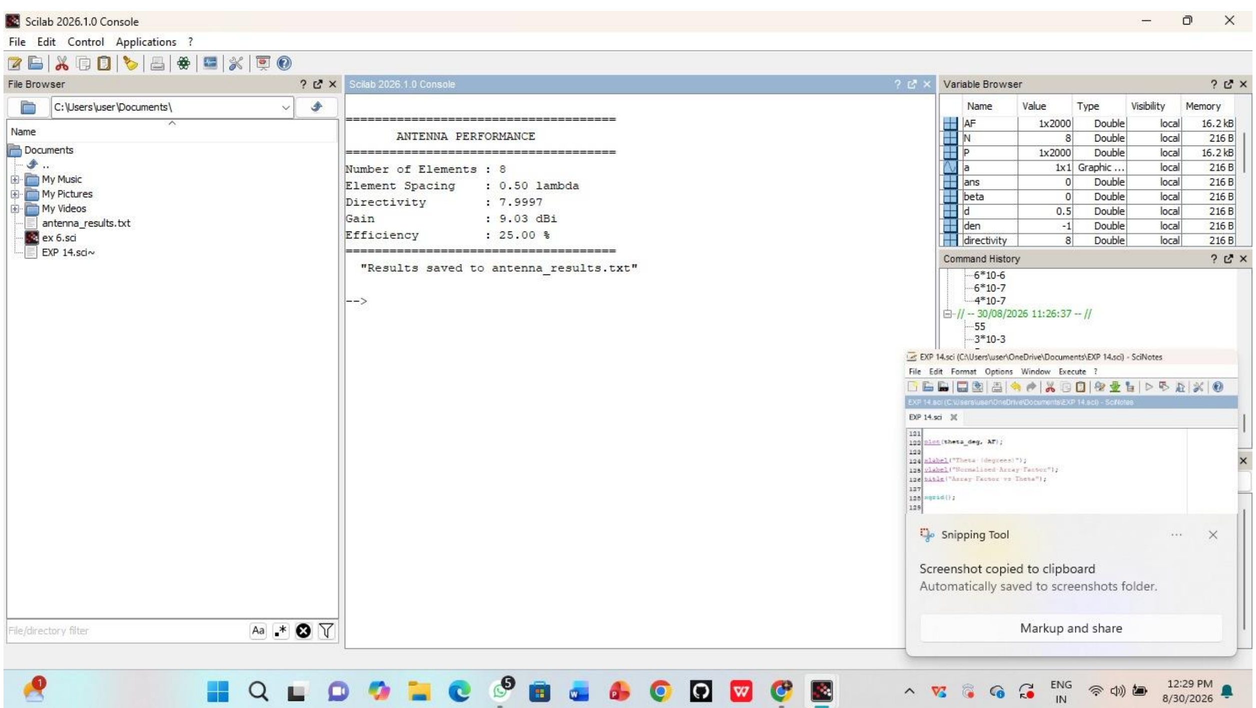
Program:

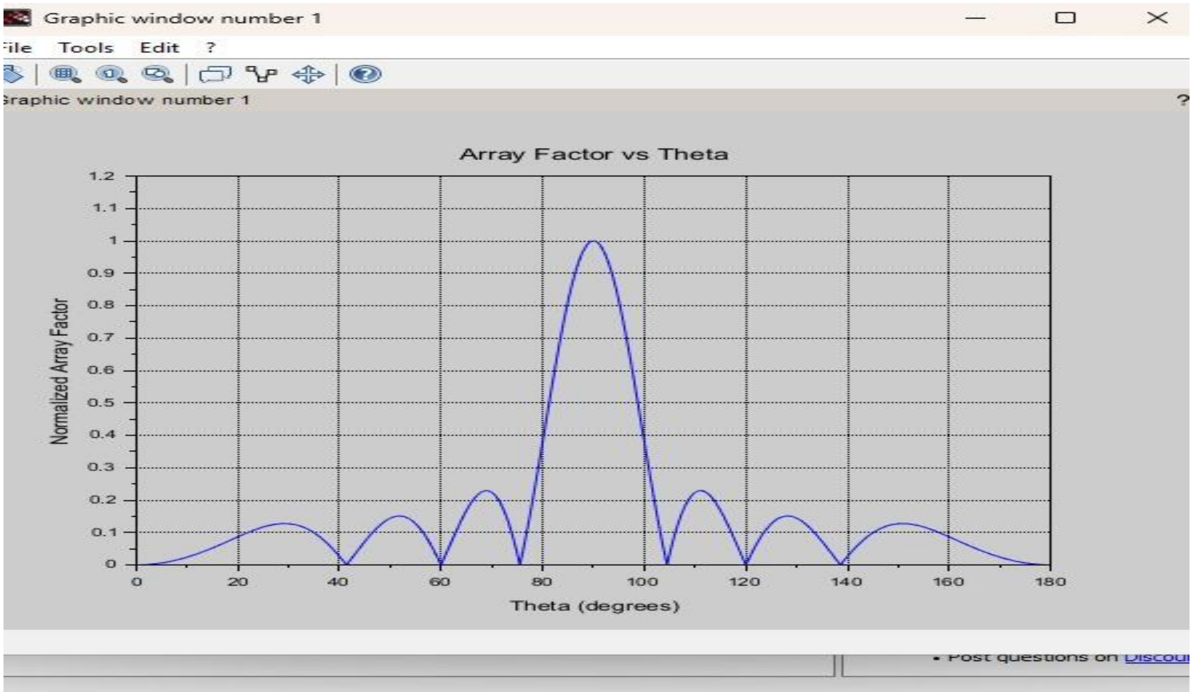

```
clc;
clear;
clf();

// ================================================
// ANTENNA EFFICIENCY ESTIMATION
// ================================================

// Antenna parameters
N = 8;                      // Number of elements
d = 0.5;                    // Element spacing in wavelength
beta = 0;                   // Phase difference

// Angular range
theta_deg = linspace(0, 180, 2000);
theta = theta_deg * %pi / 180;


// ================================================
// ARRAY FACTOR
// ================================================

psi = %pi * cos(theta) + beta;

AF = zeros(1, length(theta));

for i = 1:length(theta)

    den = sin(psi(i) / 2);

    if abs(den) < 1e-10 then
        AF(i) = 1;
    else
        AF(i) = abs(sin(N * psi(i) / 2) / ...
                    (N * den));
    end

end

// Normalize
AF = AF / max(AF);
```



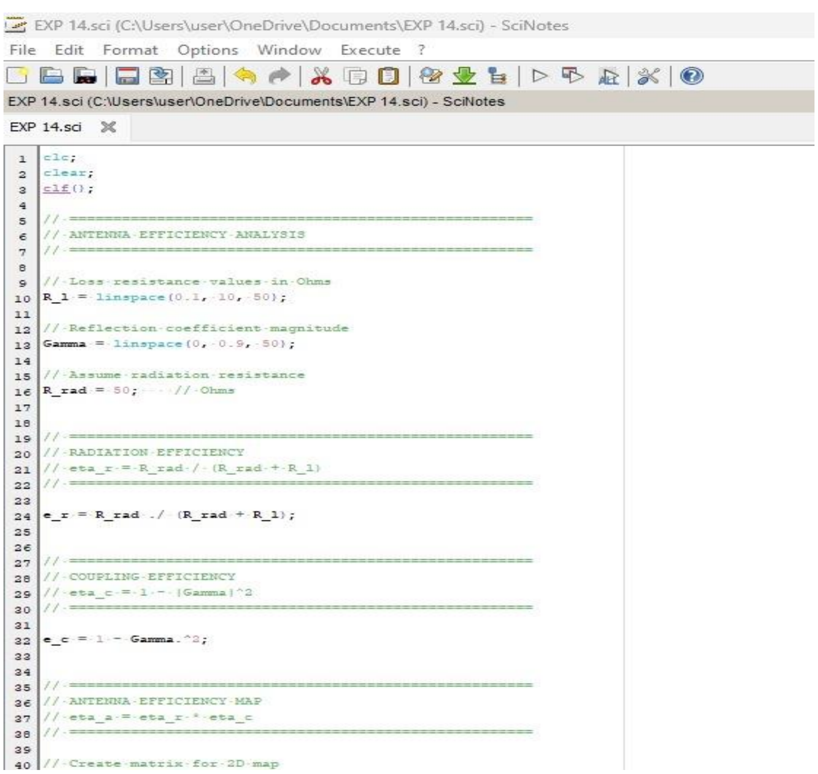
```
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

plot(theta_deg, AF);

xlabel("Theta (degrees)");
ylabel("Normalised Array Factor");
title("Array Factor vs Theta");

xgrid();

a = gca();
a.data_bounds = [0, 0; 180, 1.05];


// ==============================================
// SAVE RESULTS TO TEXT FILE
// ==============================================

filename = "antenna_results.txt";

fd = mopen(filename, "wt");

if fd == -1 then

    disp("Error: Could not create output file.");

else

    mfprintf(fd, "ANTENNA PERFORMANCE RESULTS\n");
    mfprintf(fd, "===========================\n");
    mfprintf(fd, "Number of Elements: %d\n", N);
    mfprintf(fd, "Element Spacing: %.2f lambda\n", d);
    mfprintf(fd, "Directivity: %.4f\n", directivity);
    mfprintf(fd, "Gain (dBi): %.2f\n", gain);
    mfprintf(fd, "Efficiency (%%): %.2f\n", efficiency);

    mclose(fd);

    disp("Results saved to antenna_results.txt");

end
```

Output:





Program:

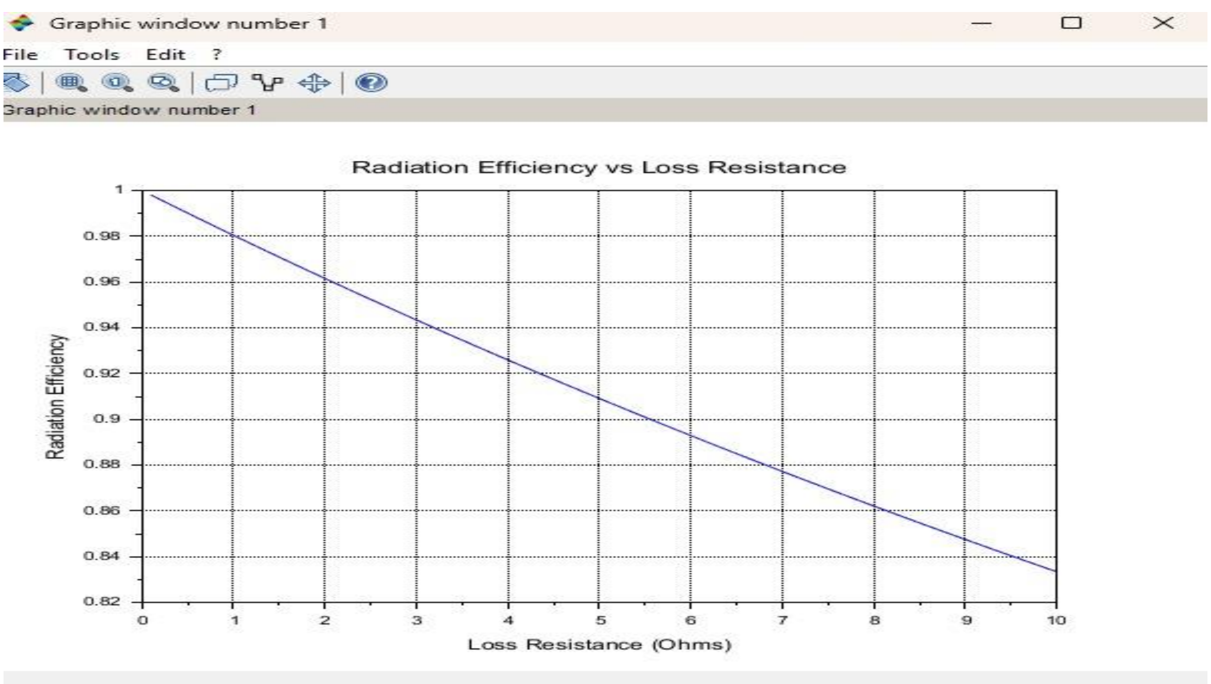


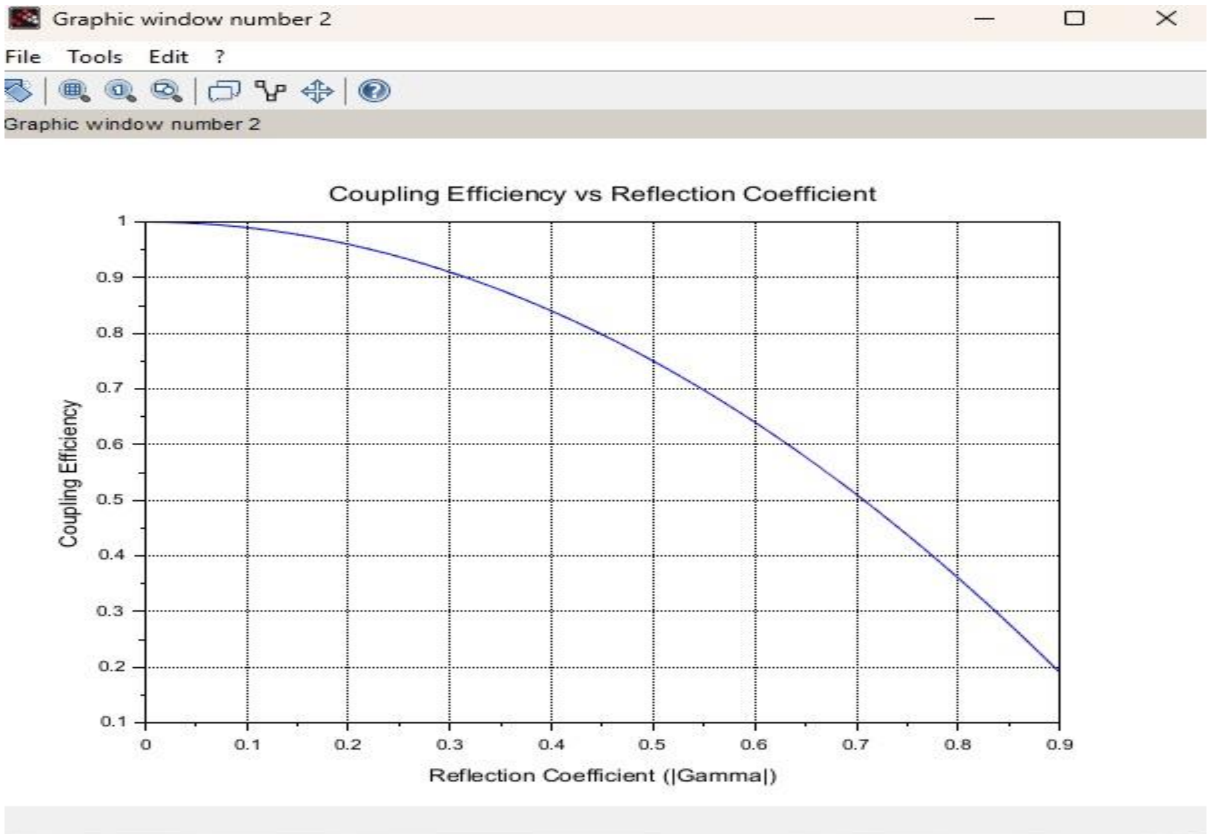

```
clc;
clear;
clf();

// =======================================================
// ANTENNA EFFICIENCY ANALYSIS
// =======================================================

// Loss resistance values in Ohms
R_l = linspace(0.1, 10, 50);

// Reflection coefficient magnitude
Gamma = linspace(0, 0.9, 50);

// Assume radiation resistance
R_rad = 50;     // Ohms


// =======================================================
// RADIATION EFFICIENCY
// eta_r = R_rad / (R_rad + R_l)
// =======================================================

e_r = R_rad ./ (R_rad + R_l);


// =======================================================
// COUPLING EFFICIENCY
// eta_c = 1 - |Gamma|^2
// =======================================================

e_c = 1 - Gamma.^2;


// =======================================================
// ANTENNA EFFICIENCY MAP
// eta_a = eta_r * eta_c
// =======================================================

// Create matrix for 2D map
```

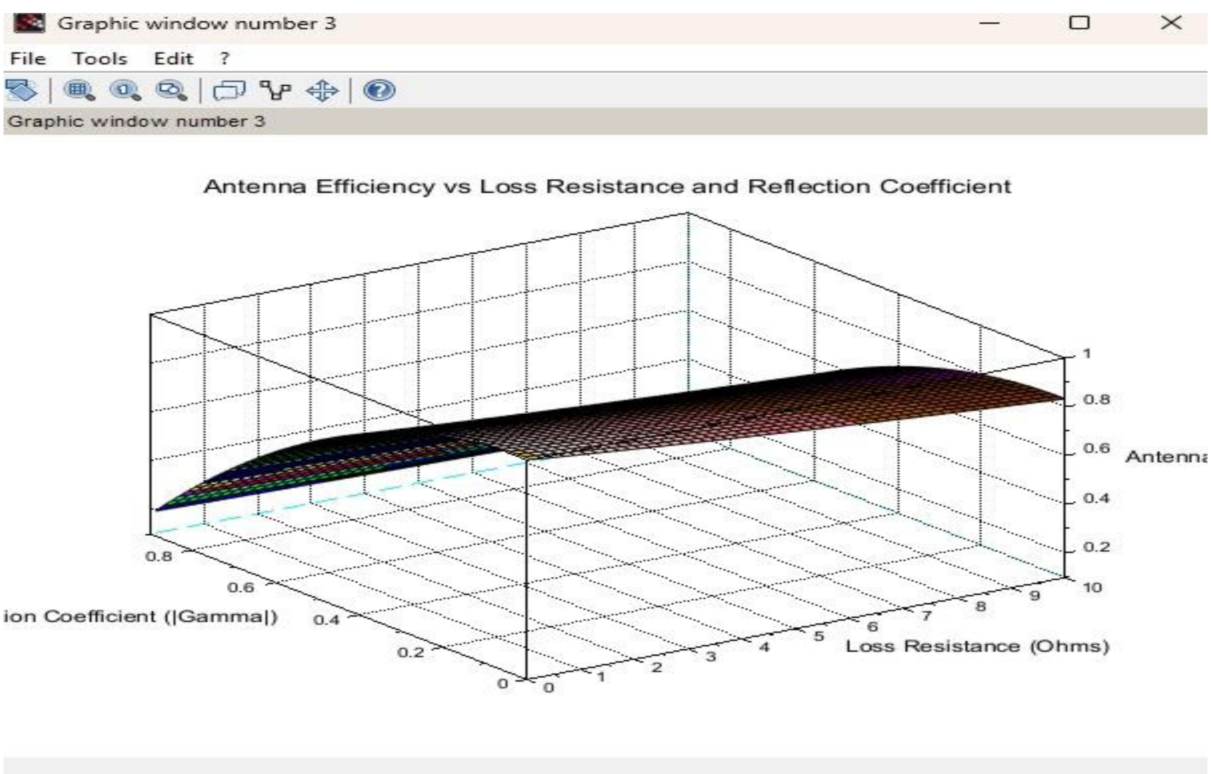
EXP 14.sci (C:\Users\user\OneDrive\Documents\EXP 14.sci) - SciNotes

```
xlabel("Loss Resistance (Ohms)");
ylabel("Radiation Efficiency");
title("Radiation Efficiency vs Loss Resistance");

xgrid();


// ============================================================
// PLOT 2: COUPLING EFFICIENCY VS REFLECTION COEFFICIENT
// ============================================================

scf(2);

plot(Gamma, e_c);

xlabel("Reflection Coefficient (|Gamma|)");
ylabel("Coupling Efficiency");
title("Coupling Efficiency vs Reflection Coefficient");

xgrid();


// ============================================================
// PLOT 3: 2D ANTENNA EFFICIENCY MAP
// ============================================================

// surf requires matrix dimensions:
// rows = Gamma
// columns = R_l

scf(3);

surf(R_l, Gamma, e_a);

xlabel("Loss Resistance (Ohms)");
ylabel("Reflection Coefficient (|Gamma|)");
zlabel("Antenna Efficiency");
title("Antenna Efficiency vs Loss Resistance and Reflection Coefficient");

xgrid();
```

Output:




Graphic window number 2
File Tools Edit ?
Graphic window number 2
Coupling Efficiency vs Reflection Coefficient
Coupling Efficiency
Reflection Coefficient (|Gamma|)
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9


Graphic window number 3
File Tools Edit ?
Graphic window number 3
Antenna Efficiency vs Loss Resistance and Reflection Coefficient
ion Coefficient (|Gamma|)
Loss Resistance (Ohms)
Antenna
0.8
0.6
0.4
0.2
0
1
0.8
0.6
0.4
0.2
0
1
2
3
4
5
6
7
8
9
10

RESULT:

The experiment was done and the efficiency of the given antenna was found. Different case studies were analyzed. These case studies highlight the practical importance of antenna efficiency in various applications. For all systems, increasing radiation efficiency (minimizing power loss) and coupling efficiency (ensuring impedance matching) were crucial for optimizing antenna performance. The calculations show that, for practical systems, small improvement